

Quality Management

4 — *What quality is, what it costs, how you organise for it*

Prof. Gianluca Rossi • dr. Lorenzo Capponi

BSc in Engineering Management • Università degli Studi di Perugia

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Today

Last time ended with a problem. Today we deal with it.

The objective moved from *conformity to specifications* to *fitness for use*. But fitness for use is judged by the customer — who is not in the room when you design the thing.

So: **how do you engineer something whose success is defined by someone else?** Three answers, in order:

1

Define it

what quality actually means once the user decides

2

Price it

what quality and its absence cost

3

Organise for it

the structures that deliver it

PART 1

What quality is

And why the producer does not get to decide

Two senses of one word

In common language

Quality as a **qualitative characteristic**, and quality as **value**. Vague, comparative, and about the object in the abstract.

In the industrial sense

Compliance with **specific measurable characteristics**, and compliance with the **use in a specific circumstance**. Precise, and about this object in this situation.

THE SAME WATCH, BOTH SENSES

“A quality watch” is the first sense — it means expensive, well-finished, admired. “Accurate to ± 2 seconds per day at 25 °C” is the second. Only the second can be measured, specified in a contract, or argued about in court.

For the rest of this course

We only use the second sense. When you say “high quality” in an exam and mean “nice”, you have said nothing.

Quality is defined by the user

Not by the producer

The working definition

The satisfaction of **expressed or implied needs**. For a service: the ability to achieve the satisfaction of the needs of users and clients.

Satisfaction is the gap between the service **as received** and the service **as imagined**. Both halves are in the customer's head. The producer controls only one of them — and only indirectly.

IMPLIED NEEDS ARE THE DANGEROUS ONES

Nobody writes “the phone must not catch fire” in a specification. Nobody asks for it. Everybody expects it. An implied need is invisible until it is violated, and then it is the only thing anyone talks about.

A CHEAP PEN CAN BE HIGH QUALITY

A €1 pen that always writes has high quality. A €300 pen that leaks in your pocket does not. Price is not quality; meeting the need is.

Expressed and implied needs

Only one of the two is written down

Expressed

Stated in the order, the specification or the contract.
Arguable, checkable, and the easy half.

Implied

Never stated, universally assumed. Invisible until violated — and then the only thing anyone discusses.

THREE IMPLIED NEEDS NOBODY WRITES DOWN

A restaurant meal will not make you ill. A lift will not move with the doors open. A phone charger will not set fire to the flat. No customer has ever asked for these, and no supplier is forgiven for missing them.

THE ENGINEERING PROBLEM

You cannot design against a requirement you have not written. So how does an organisation systematically discover its implied requirements *before* a customer discovers them for it?

Five kinds of quality in a service

The gaps between them are where complaints come from

- **Expected** — analysis of the needs and expectations of users and clients.
- **Designed** — the qualitative and quantitative design of the system that delivers the service, and the definition of operating standards.
- **Delivered** — what the service actually was, on the day.
- **Perceived** — control and detection of the degree of satisfaction of users and clients.
- **Compared** — the result of comparison with other similar services, to identify differences in quality.

WHERE THE HOTEL LOSES THE CUSTOMER

Expected: someone at reception at 2 a.m. **Designed:** a night bell. **Delivered:** a bell nobody answers.

Perceived: “they don’t care.” **Compared:** the hotel across the road has a night porter.

Four separate failures, four separate fixes. Lumping them together as “bad service” tells you nothing about which one to repair.

What quality is not

Quality is not:

- something you add to your work only when you have spare time
- keeping information to yourself instead of sharing aims
- a lack of professionalism

And three things to hold onto:

- quality is a **business objective**, not perfection
- quality is **relative**, not absolute
- quality is the **expected** product or service, not the perfect one

Quality is the compromise of three needs

Effectiveness (quality of product or service) • **Efficiency** (costs and timing) • **Elasticity** (ability to absorb modifications)

WHY “RELATIVE, NOT ABSOLUTE” MATTERS

A surgical implant and a disposable fork are both allowed to be high quality. They are judged against different expected uses. Engineering the fork to implant standards is not excellence — it is waste.

Quality is a compromise of three needs

Push any one to its limit and you destroy the other two

Effectiveness

Quality of the product or service — does it do the job?

Efficiency

Costs and timing — at what price, and how fast?

Elasticity

Ability to absorb modifications — what happens when the requirement changes?

WATCH WHAT HAPPENS AT THE EXTREMES

Maximum effectiveness alone gives you a product nobody can afford. Maximum efficiency alone gives you the cheapest thing that technically qualifies. Maximum elasticity alone gives you a design so general it is good at nothing. The engineering is in the compromise, not in any one axis.

APPLY IT

Take a product you know well. Which of the three did its designers prioritise, and what did they visibly give up to get it?

- **1 Two senses of the word.** Common language means value and taste; the industrial sense means measurable characteristics and fitness for a specific use. We only use the second.
- **2 The user defines quality,** through expressed *and* implied needs. The implied ones are the dangerous ones, because you cannot design against a requirement nobody wrote.
- **3 A service has five qualities** — expected, designed, delivered, perceived, compared. Complaints live in the gaps between them, and each gap has a different fix.
- **4 Quality is a business objective, relative, and expected** — not perfection, not absolute, not perfect. It is the compromise of effectiveness, efficiency and elasticity.

PART 2

What quality costs

*The argument that gets quality onto the board's
agenda*

The costs of quality

Handley, 2000

$$\text{CoQ} = \text{PoC} + \text{PoNC}$$

PoC — Price of Conformance

What it costs you to do it the proper way from the beginning: prevention, training, design reviews, calibration, inspection.

PoNC — Price of Non-Conformance

What it costs you because you did not: scrap, rework, warranty, recalls, lost customers, the engineer's week spent on the complaint.

The number that ends the argument

The costs of **non-quality** run at roughly **25–40% of total costs**. That is not a quality department budget. That is a quarter to two-fifths of everything the company spends.

THE 1–10–100 RULE OF THUMB

A defect costs €1 to prevent at design, €10 to catch on the line, €100 once the customer has it. The numbers are illustrative; the orders of magnitude are not.

Where the money actually sits

Four categories — the standard way to split the cost of quality

Prevention — stopping it happening

Training, design review, process planning, supplier qualification, calibration programmes.

Appraisal — finding out whether it happened

Inspection, testing, audits, measurement, laboratory work.

Internal failure — caught before it shipped

Scrap, rework, re-inspection, downgrading, downtime.

External failure — caught by the customer

Warranty, returns, recalls, complaint handling, liability, lost customers.

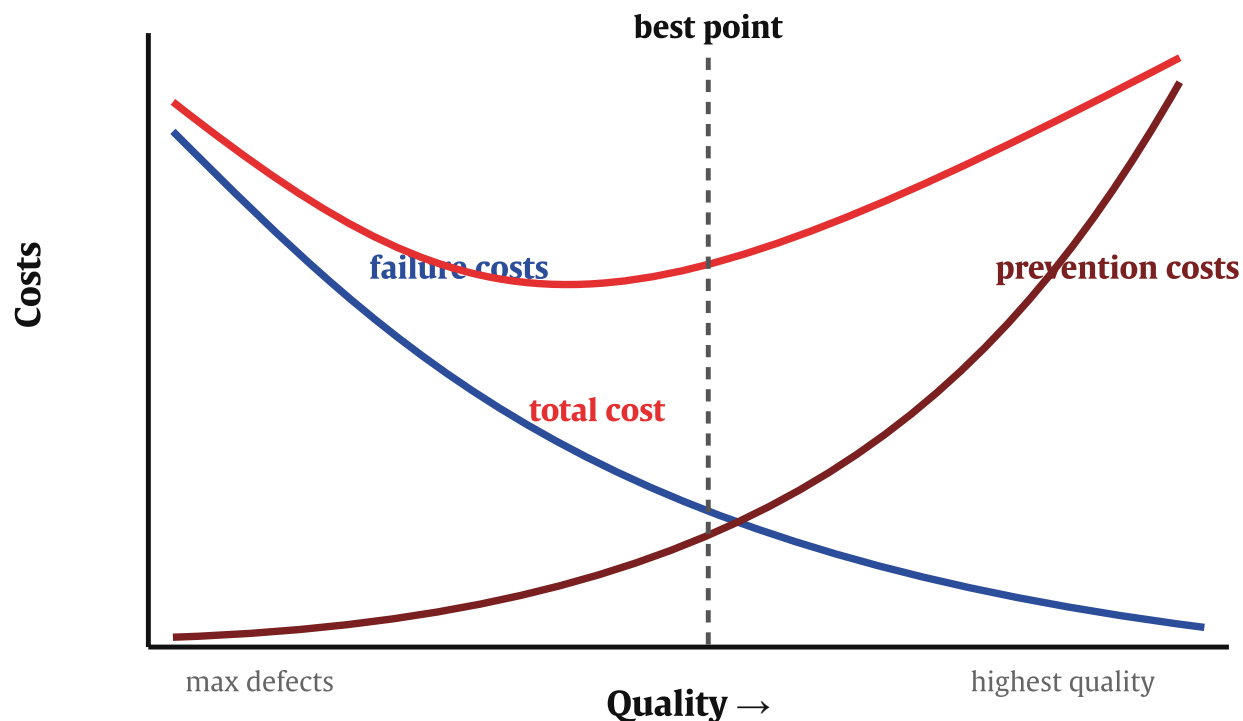
The first two together are the **price of conformance**. The last two are the **price of non-conformance**.

THE RATIO TELLS YOU WHICH COMPANY YOU ARE LOOKING AT

A company spending heavily on *appraisal* and *internal failure* is inspecting quality in — it is the 1950s. A company whose spend has shifted to *prevention* has moved upstream. Same total, completely different organisation.

Is there an optimum?

The classical picture — and the objection to it



As quality rises, failure costs fall and prevention costs rise. The total has a minimum somewhere short of perfection — the **best point of production**.

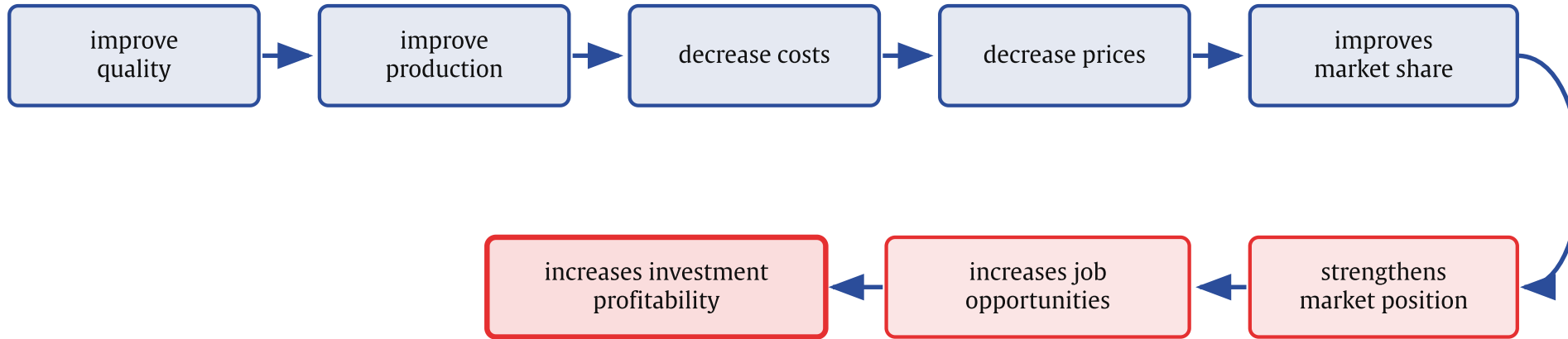
The objection

Deming and Juran argued this curve is drawn wrong. Prevention costs do *not* rise without limit: as a process improves, prevention gets **cheaper**, because you stop needing the inspection. On that view the optimum keeps sliding right, towards zero defects.

Both models are defensible. Know which one you are arguing from.

Deming's chain reaction

Why improving quality is not a cost centre



THE COUNTER-INTUITIVE STEP

Improve quality → decrease costs. Everyone expects the opposite. It works because you stop paying for scrap, rework, inspection and warranty — and those were the 25–40%.

What quality costs

- **1** **CoQ = PoC + PoNC.** Prevention and appraisal are the price of conformance; internal and external failure are the price of non-conformance.
- **2** **Non-quality runs at 25–40% of total costs.** That is not a department budget — it is a quarter to two-fifths of everything the company spends.
- **3** **Cost escalates by roughly an order of magnitude at each stage** — prevent for €1, catch on the line for €10, let the customer find it for €100.
- **4** **The classical optimum curve is disputed.** Deming and Juran argued prevention gets cheaper as the process improves, pushing the optimum towards zero defects.
- **5** **Improving quality lowers cost,** because the cost was sitting in the failures. That is Deming's chain reaction, and it is the argument that wins a budget.

PART 3

Organising for quality

Structures, tools, and the cycle that ties them together

The market presents itself in two ways

The customer requests a specific product

The company produces **on order**. There is a contractual relationship between supplier and customer, and the specifications are written down.

The customer chooses from the market

The company produces a **series product**. Nobody negotiates anything. The customer either buys it or does not.

The obligation is the same in both cases

Satisfying the customer means satisfying the **expressed** specifications *and* the **implicit** ones. The second case does not let you off — it only removes the document you could have pointed at.

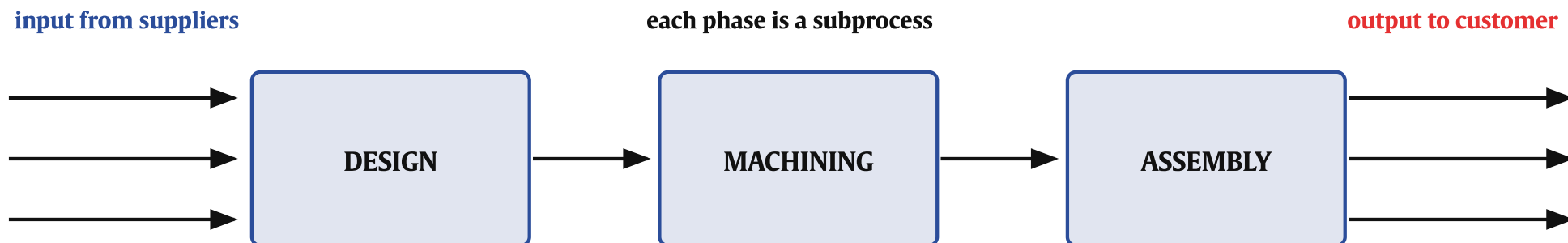
For mass-market products the company must define specifications for needs the customer never expressed, across parameters perceived collectively: **reliability • safety • aesthetics • performance • cost** — and translate them into specifications compatible with costs.

CUSTOM MACHINE TOOL VS WASHING MACHINE

For the machine tool you argue about tolerances in a meeting. For the washing machine you discover what people implicitly expected when the reviews appear.

Everyone is a customer of someone

The client–supplier relationship, applied inside the company



Quality systems, according to ISO 9000, are built on the **client–supplier relationship**. Each phase of production is a subprocess in itself, and the quality of the whole depends on the quality of every phase.

THE DRAWING OFFICE HAS CUSTOMERS

The machine shop is the drawing office's customer. A dimension that is ambiguous on the drawing is a *defective delivery* — even though no product has been made yet, and nobody outside the building will ever see it.

The five zeros

The operational targets of total quality

- **Zero defects** — a perfect product or service, with no errors or mistakes.
- **Zero inventory** — no excess stock or materials, reducing cost and improving efficiency.
- **Zero setup time** — minimising the time needed to change from producing one item to another.
- **Zero stoppages** — eliminating interruptions and downtime in the production process.
- **Zero paper** — a paperless office and production environment.

ZERO SETUP TIME IS THE ONE THAT UNLOCKS THE OTHERS

Toyota reduced a die change from hours to under ten minutes. Once changeover is cheap, you can make small batches — which lets you hold almost no stock, and means a defect is caught after ten pieces instead of ten thousand. The five zeros are not five separate goals; they hold each other up.

Two engines, pushing in opposite directions

A company needs both

Quality circles — BOTTOM-UP

Japan, late 1960s. Around ten people from the same sector, with their supervisor, meeting regularly and voluntarily to identify, analyse and propose solutions to problems visible to operational staff but not to management. Led by an internal leader, *not* a boss. Part of CWQC.

Tools: data collection, histograms, cause-and-effect diagrams, Pareto analysis, stratification, control charts, correlation diagrams.

Risks: raising problems without proposing solutions, fruitless criticism, questioning management and company policy.

Quality committee — TOP-DOWN

A vertical structure responsible for governing quality issues, where decisions are made by top management. It involves the heads of all departments: sales, design and R&D, quality, procurement, organisation, production, marketing, after-sales.

Unlike circles, it presents **solutions** to top management, and its analyses must be communicated **in economic terms**, so that decisions rest on quantitative economic parameters.

Why both

Circles see what management cannot. The committee can authorise what circles cannot. A circle that finds a €200,000 problem still needs someone able to sign for it.

Quality circles, in detail

Japan, late 1960s — and why they are harder than they look

How they are constituted

- around ten people, from the same sector, with similar activities
- participation is **spontaneous and voluntary**, open to everyone
- led by an internal leader using charisma — *not* a boss issuing instructions
- results are public and disseminated, which drives emulation and awareness

What they are for

- **Corporate:** stimulate work, use the individual experience of the workforce, improve conditions, enhance product quality
- **Individual:** involve staff in management activities, motivate improvement of their own environment, encourage focus on quality, humanise the relationship with management

WHY “VOLUNTARY” IS LOAD-BEARING

A mandatory quality circle is a meeting. The value comes from people choosing to spend their own attention on a problem management had not noticed — and that choice cannot be ordered.

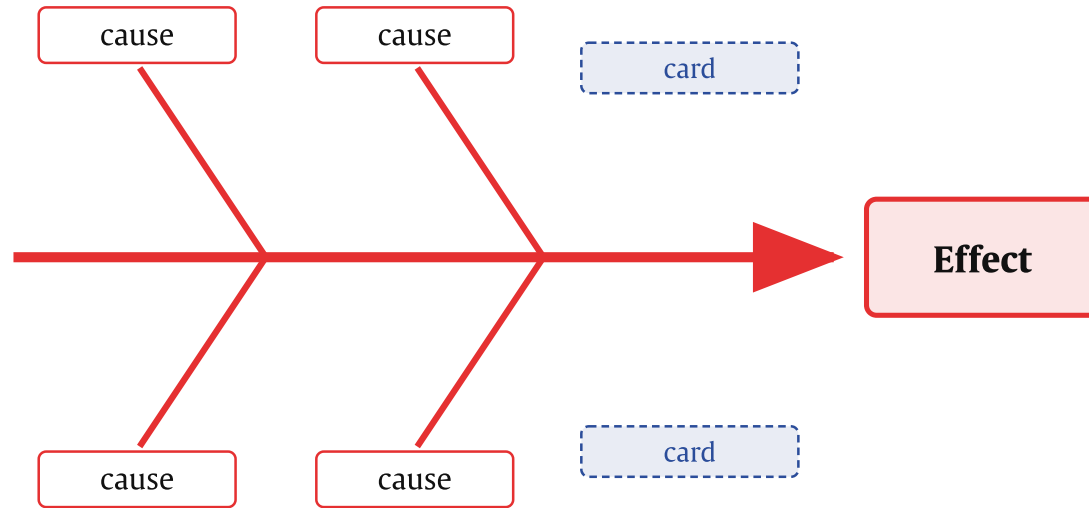
The seven tools

What a circle actually uses — and a preview of the statistics block

- **Data collection and processing** — the check sheet. Before anything else, count something.
- **Histograms** — what the distribution of a measurement actually looks like.
- **Cause-and-effect diagrams** — Ishikawa's fishbone, for organising hypotheses about causes.
- **Pareto analysis** — Juran's "few essentials, many modest", made into a chart.
- **Stratification** — splitting data by shift, machine, operator or batch, so an average stops hiding the answer.
- **Control charts** — Shewhart's tool: is this variation worth chasing, or is it the process being itself?
- **Correlation diagrams** — does this variable actually move with that one?

Note what these have in common

Every one is simple enough to be used by someone who is not a statistician, on the shop floor, with a pencil. That was the design requirement — and it is why they spread.



The diagram is posted in a visible place. Anyone can add a card.

Five phases:

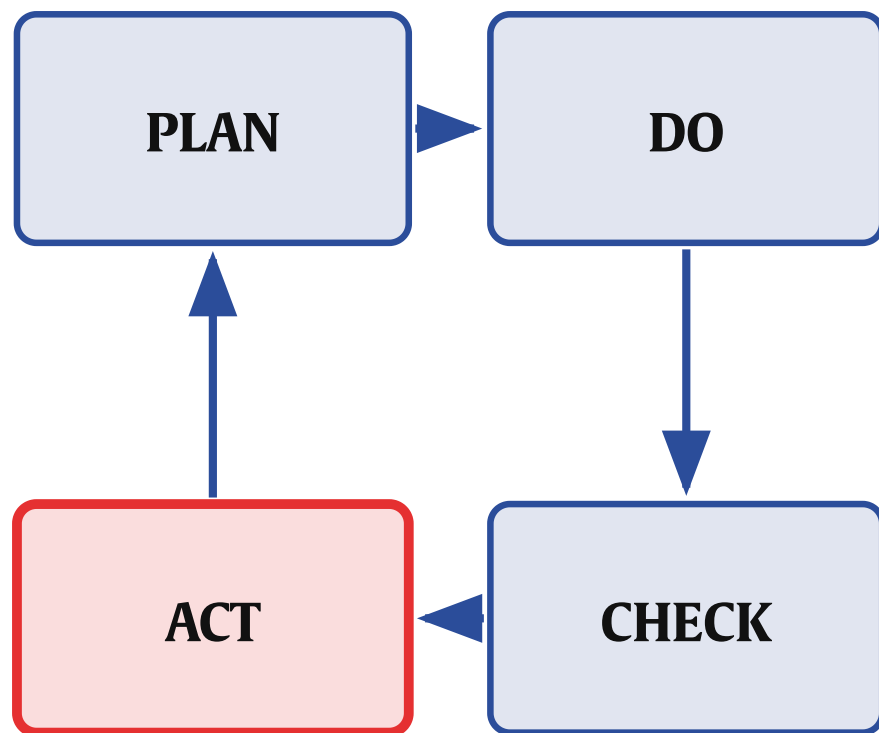
- Identify an **important** problem and set a quantitative goal — a trivial first problem discredits the method
- List the factors and build the cause-and-effect diagram, with workers and engineers
- Prioritise causes by their impact, using statistical tools
- Develop solutions — feasibility, cost, effectiveness
- Implement and follow up; standardise what works

Root causes fall into four categories

lack of knowledge • incomplete transmission of knowledge • incorrect implementation of knowledge • lack of production methods and technologies

PDCA

The cycle everything else hangs on



and round again, at a higher level

- **Plan** — define new quality objectives
- **Do** — design and implement the interventions
- **Check** — verify the results
- **Act** — take action within the company to consolidate the results, and restart towards new objectives

ACT IS THE STEP EVERYONE SKIPS

Most organisations plan, do, and check — then move on to the next fire. Without ACT the improvement is never consolidated, the old habit returns, and the same problem is rediscovered next year by someone else.

The quality manual

Where all of this ends up written down

It contains all the basic indications for all the activities, and points at three collections of procedures:

Management procedures

the operational processes • specific individual and interface activities • the management • the documents to issue and manage

Technical procedures

the specific process or project activities • the applicable regulations • the necessary checks and when they are carried out • the certificates to be issued

Quality control procedures

the methods for carrying out the checks
• the methods for issuing the results of the checks and the certificates

The failure mode

This is exactly where the bureaucratic degeneration we saw in TQC begins. A manual nobody reads is not a quality system — it is a filing cabinet.

Conclusions

- Quality is a **business objective**, not perfection; **relative**, not absolute; the **expected** product, not the perfect one
- It is defined by the **user** — through expressed and implied needs alike
- **CoQ = PoC + PoNC**, and non-conformance runs at 25–40% of total costs. That is the argument that gets quality a budget.
- Improving quality *lowers* cost, because the cost was in the failures
- Inside the company everyone is a customer of someone — the client–supplier relation applies to every phase
- Circles push up, the committee pushes down, and **PDCA** keeps both turning

Next time

ISO 9000 — what happens when all of this is written into a standard you can be certified against.

Questions?

Prof. Gianluca Rossi • dr. Lorenzo Capponi

`lorenzo.capponi@fs.uni-lj.si`

`github.com/LolloCappo/student-hub`